Decomprolute: a proteomic tumor deconvolution benchmarking framework

Song Feng[1], Anna Calinawan[2], Francesca Petralia[2], Pei Wang[2], Pietro Pugliese[3], Michele Ceccarelli[4], Sara Gosline[1]

1. Pacific Northwest National Laboratory, Seattle, WA, 2. Icahn School of Medicine at Mount Sinai, New York, NY; 3. University of Sannio, Italy; 4. University of Naples "Federico II", Italy.

Unmet need for deconvolution benchmarks

Tumor deconvolution algorithms[1-4] have become a reliable way to disentangle the diverse cell types that comprise solid tumors. In the absence of single cell data, these **algorithms** use distinct sets of genes, **signature matrices,** that represent distinct cell types to estimate the fraction of cell types observed in each tumor. The development of gold standards and platforms for RNA-based deconvolution has enabled the development of robust tumor deconvolution algorithms that are employed across RNA-seq datasets.

Through the efforts of the pan-cancer working groups of the Clinical Proteomics Tumor Analysis Consortium we have now seen that proteomics data provides distinct insights into tumor biology compared to mRNA, and believe that the data can be used to disentangle the complicated cellular heterogeneity of tumors. To date, however, there are neither software algorithms designed to deconvolve proteomic data nor are there gold standard datasets by which to measure the performance of these algorithms.

To address this, we introduce Decomprolute, a platform for the benchmarking of tumor deconvolution algorithms designed to support the development of novel algorithms and gold standards for proteomic deconvolution. We designed three independent metrics that we hope will encourage further development in the field.

Metric 1: Comparison of matched mRNA and protein samples

There are numerous algorithms and 'gold standard' datasets for the deconvolution of tumor samples from patient-derived mRNA including Cibersort[1], MCPCounter[2], XCell[3], and EPIC[4] (four of the algorithms we included in our platform). As such, we leveraged the nearly 1000 matched mRNA-protein patient samples available across 10 different cancer types to evaluate how similarly the mRNA and protein predictions are to one another.

Here we iterate over each algorithm and cell signature matrix and calculate the Spearman rank correlation to assess the correlation across cell type and patient between cell type predictions made on mRNA vs. protein. When run, the workflow outputs results from this analysis depicted in **Figure 2**. We see a high correlation between xcell and mcpcounter predictions across cancer type and signature matrix (**Figure 2A**). **Figure 2B** shows the breakdown by cell type for the LM7c signature matrix. This shows the consistent correlation between xcell and mcpcounter and also highlights the poor agreement with epic mRNA-based predictions.

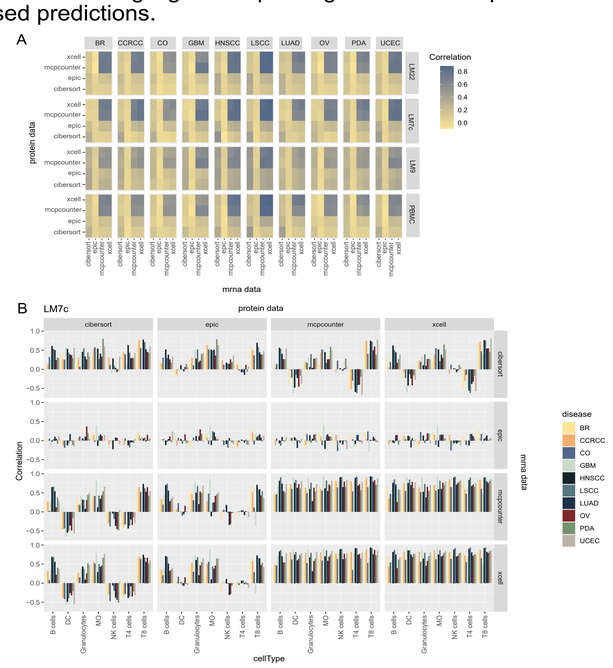

Figure 2: Standardized decomprolute output depicts correlation between mRNA and proteomic deconvolution predictions on matched patient samples. (A) Heatmap panels depicting mean rank correlation between mRNA (x axis) and protein (y-axis) based predictions for each algorithm, cancer type, and signature matrix. Legend to the right. (B) depicts mean correlation between mRNA (rows) and protein predictions (columns) across the cell types for the LM7c signature matrix. Average values of each cancer type are represented by a different color.

Modular framework benchmarks proteomic tumor deconvolution

We containerized the CPTAC data, standard deconvolution algorithms, and signature matrices as depicted in **Figure 1** below. We then built Common Workflow Language (CWL)[5] scripts to coordinate the execution of the deconvolution algorithms across the different data sets and signature matrices to compare their performance using three individual metrics. All code is at **https://github.com/pnnl-compbio/decomprolute** The results of the metrics are depicted in Figures 2-4.

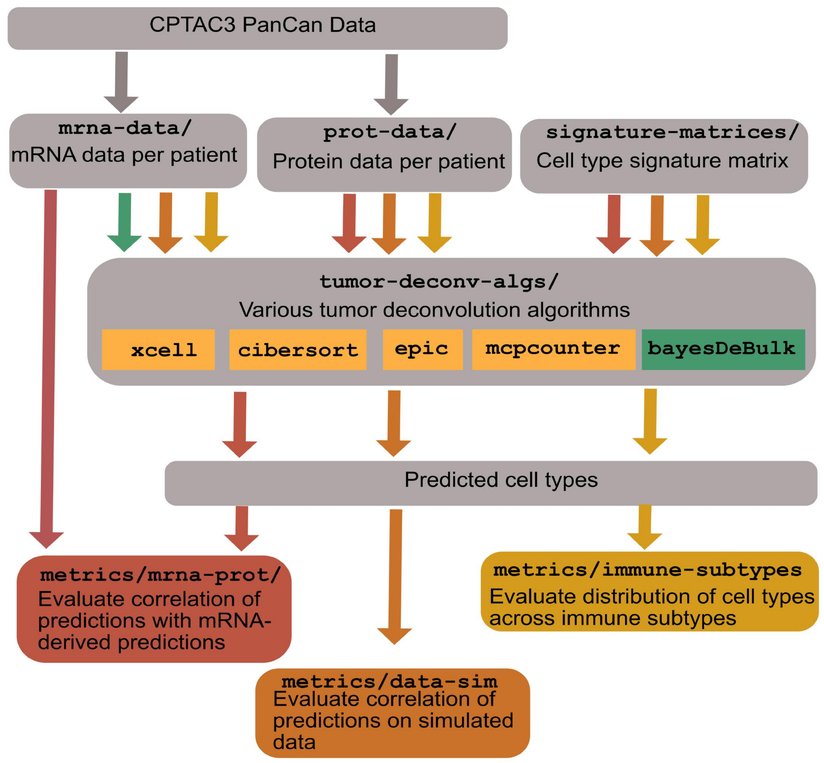

Figure 1: Architecture of containerized CWL workflow. CPTAC data is downloaded via the cptac Python package for both mRNA and proteomic data types into a individual Docker containers. Four signature matrices are captured in a third Docker container and four deconvolution algorithms are each in their own container. The three metrics are also containerized as well as the figure generation.

Metric 2: Algorithmic performance on simulated datasets

To compare matrices, we simulated data from each matrix using a fixed estimation of cell types from sorted proteomics data[7]. We then computed the correlation between the simulated cell types and those predicted from the algorithms. A comparison of two simulations (**Figure 3**), show that LM7c is able to predict granulocytes more accurately by combining basophil, eosinopil and neutrophil predictions.

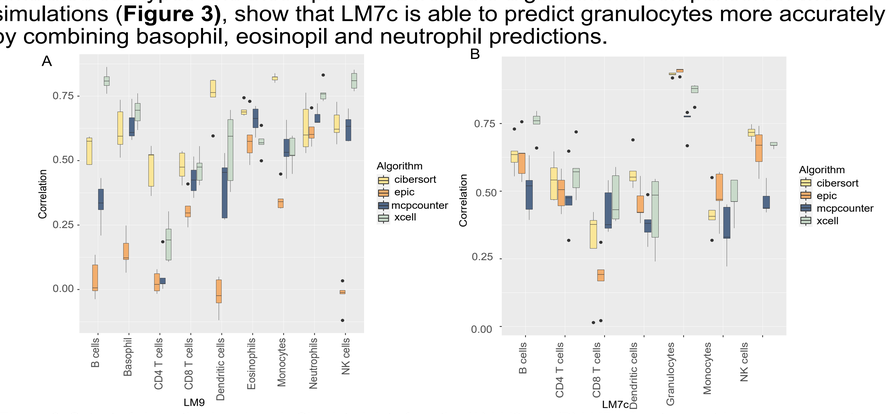

Figure 3: Output of simulation benchmark. Simulated data from the (A) LM9 and (B) LM7c signature matrix were compared to predictions made on fixed cell type populations derived from 7. Y-axis represents correlation between estimate cell types and simulated, while x-axis represents distinct cell types.

Common workflow language (CWL) benchmark scripts

We used the common workflow language (CWL) to orchestrate the benchmark metrics and enable re-use. This requires only a CWL implementation and Docker.[6] More details at: **https://pnnl-compbio.github.io/decomprolute/**

You can use the framework to deconvolve a single dataset as below:

```
cwltool metrics/prot-deconv.cwl --protAlg xc
--signature LM22 --cancerType BRCA --sampleType all
```

Or you can run the metrics evaluation below to achieve results in Figure 2, Figure 3, and Figure 4:

mRNA-protein comparison (Metric 1):

```
cwltool mrna-prot/mrna-prot-comparison.cwl
mrna-prot/pan-can-matrix.yml
```

Simulated comparison (Metric 2):

```
cwltool data-sim/simul-data/comparison.cwl
data-sim/prot-sim-test.yml
```

Immune subtype comparison (Metric 3):

```
cwltool imm-subtypes/pan-can-immune-preds.cwl
imm-subtypes/imm-args.yml
```

Metric 3: Cell type prediction across immune subtypes

The third benchmark leveraged the matched mRNA data for each patient to classify each sample by one of the 6 immune subtypes[8] shown in **Figure 4A**. This shows how some samples will be expected to have a strong immune infiltration compared to others. We then plotted the distribution of cell type predictions across these patients, shown in **Figure 4B**. Here we see that some algorithms, such as xcell, are able to show enrichment of immune cells in the C6, C2, or C3 subtypes, while other algorithms suggest a similar distribution of cell types.

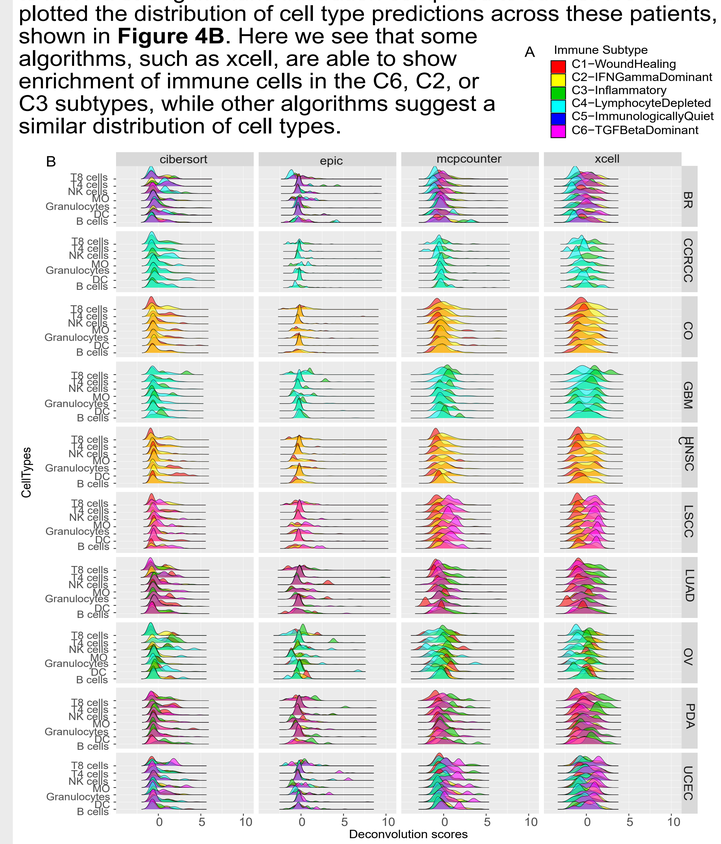

Figure 4: Distribution of cell type predictions from LM7c matrix by predicted immune subtype (as defined by mRNA). Some algorithms show better separation of cell types across immune subtypes than others.

References: 1. Newman, Aaron M., et al. "Robust enumeration of cell subsets from tissue expression profiles." Nature methods 12.5 (2015): 453-457. 2. Becht, Etienne, et al. "Estimating the population abundance of tissue-infiltrating immune and stromal cell populations using gene expression." Genome biology 17.1 (2016): 1-20. 3. Aran, Dvir, Zicheng Hu, and Atul J. Butte. "xCell: digitally portraying the tissue cellular heterogeneity landscape." Genome biology 18.1 (2017): 1-14. 4. Racle, Julien, and David Gfeller. "EPIC: a tool to estimate the proportions of different cell types from bulk gene expression data." Bioinformatics for Cancer Immunotherapy. Humana, New York, NY, 2020. 233-248. 5. Amstutz, Peter, et al. "Common workflow language, v1. 0." (2016). 6. http://docker.com 7. Rieckmann, Jan C., et al. "Social network architecture of human immune cells unveiled by quantitative proteomics." Nature immunology 18.5 (2017): 583-593. 8. Thorsson, Vésteinn, et al. "The immune landscape of cancer." Immunity 48.4 (2018): 812-830.

SCAN ME